RTL-SDR Narrow FM (NFM) Demodulator

Introduction

This Jupyter notebook implements a **complete Narrow FM (NFM) demodulation pipeline** using a raw IQ sample file captured via an RTL-SDR dongle. The goal is to receive, process, and decode a narrowband FM signal — in this case tuned to **144.470 MHz** (2m amateur band) — and produce a playable audio output.

The primary motivation to put this notebook together was my personal curiosity in how FM demodulation with SDR receivers works in detail, but after it was finished I thought it might be interesting for other people too.

What is Narrow FM?

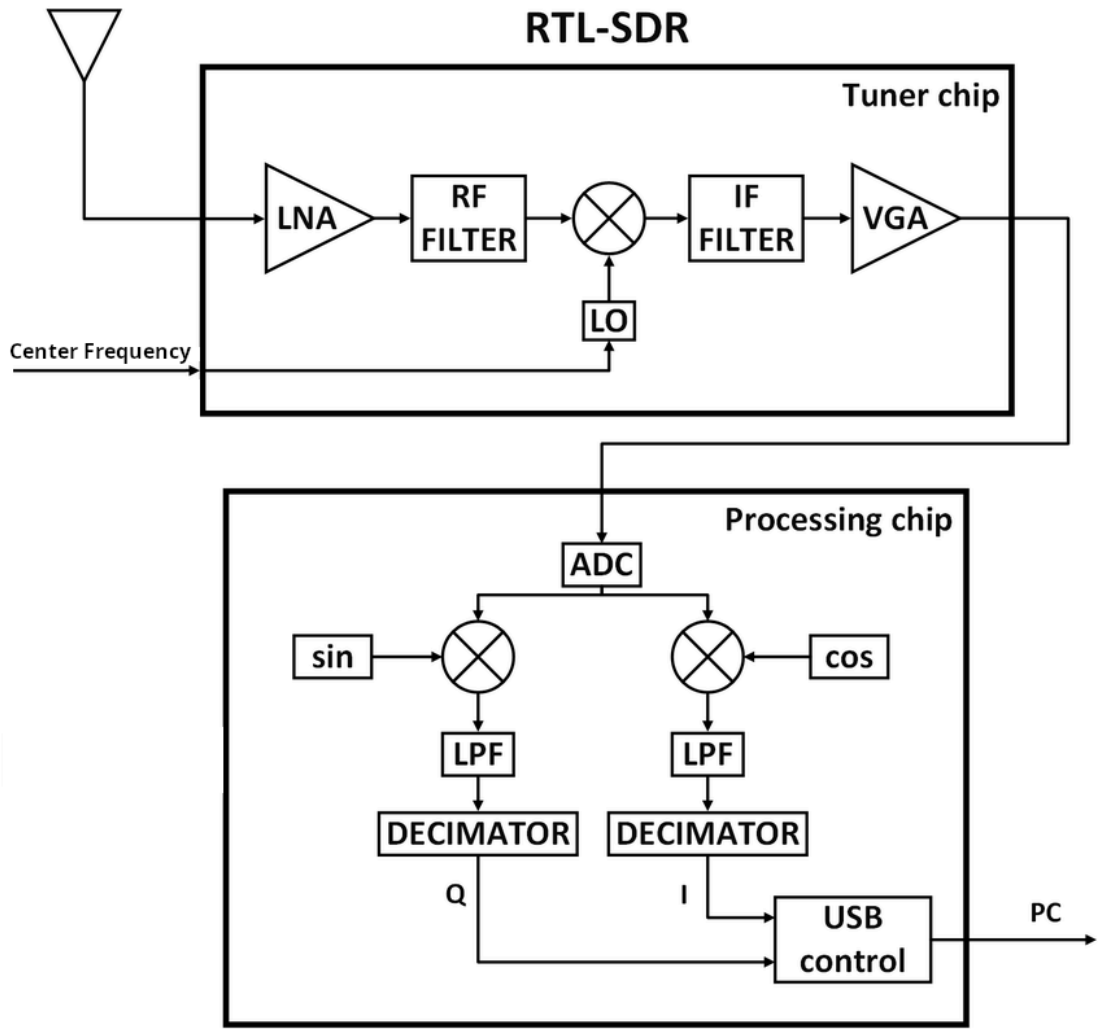
Narrow FM is a frequency modulation scheme with a small frequency deviation (typically ±2.5 kHz), commonly used in:

- Amateur radio (VHF/UHF bands)
- PMR446 / business-band radio
- Emergency services communications

It differs from wideband FM (broadcast radio) primarily in its tight channel bandwidth (~8.5 kHz IF vs. ~200 kHz for broadcast), demanding precise filtering and careful gain management to extract intelligible audio.

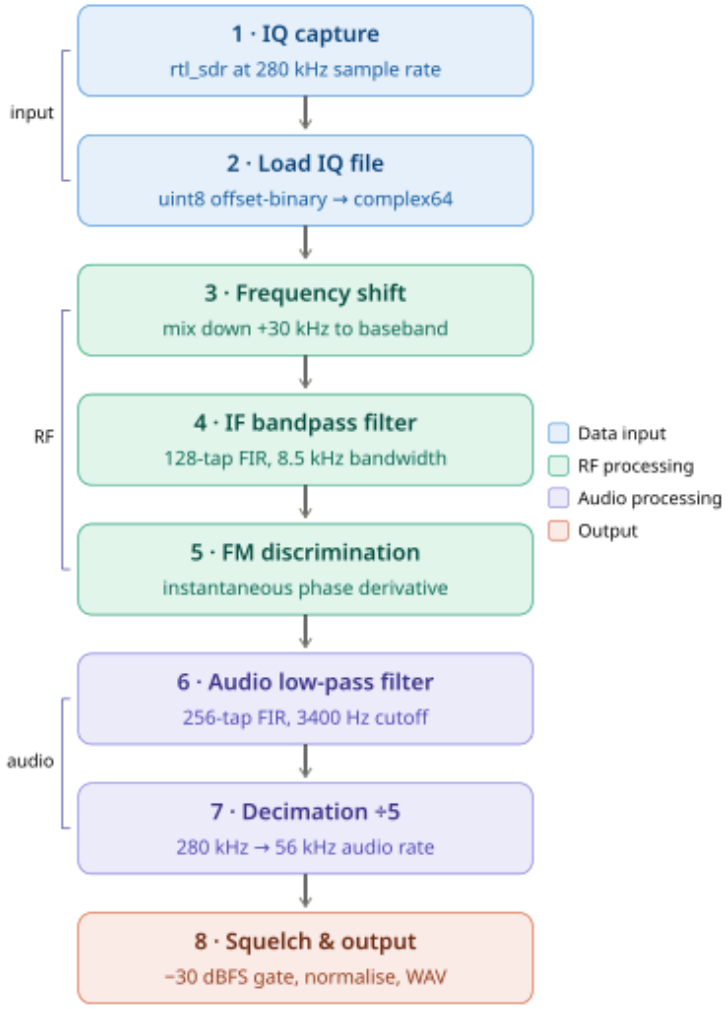
A Little Background on RTL-SDR Dongles

The RTL-SDR (Realtek Software Defined Radio) dongle is a low-cost, versatile hardware device that transforms a personal computer into a powerful software-defined radio receiver. Originally designed as a consumer DVB-T (Digital Video Broadcasting – Terrestrial) TV tuner, it was discovered that its integrated RTL2832U controller chip could be repurposed to output raw, unprocessed I/Q (In-phase/Quadrature) data. This capability, combined with a wide tuning range (typically 24 MHz to 1.7 GHz), makes it the most popular entry-level platform for hobbyists, educators, and professionals to explore the radio frequency (RF) spectrum. From listening to aircraft (ADS-B), amateur radio, and public safety channels to monitoring weather satellites and analyzing wireless protocols, the RTL-SDR has democratised access to radio technology.



Pipeline Overview

The demodulation chain implemented here consists of the following steps:



Key Parameters

```
CENTER_FREQ   = 144_470_000  # Hz  — SDR tuned frequency
SIGNAL_FREQ   = 144_500_000  # Hz  — actual signal frequency
SAMPLE_RATE   =     280_000  # Hz  — IQ sample rate
AUDIO_RATE    =      48_000  # output audio sample rate (Hz)
NFM_DEVIATION =       2_500  # Hz  — peak frequency deviation
NFM_BANDWIDTH =       8_500  # Hz  — IF channel bandwidth
AUDIO_CUTOFF  =       3_400  # Hz  — audio low-pass cutoff
```

Dependencies

Install all required packages with:

```
pip install numpy scipy matplotlib sounddevice jupyter ipywidgets
```

Capture IQ samples with:

```
rtl_sdr -f 144470000 -s 280000 -d0 -g 4 -n 1728000 capture.iq
```

The samples number 1728000 was calculated by multiplying the sample rate 280000 samples/s with the intended recording time of 6 seconds.

Output

A 6-second decoded audio file (`output.wav`) and 11 diagnostic plots covering every stage of the pipeline — from raw IQ to the final audio waveform.

The execution of the entire Jupyter Notebook can take up to 3 min.

Using a Different IQ Samples File

The provided IQ samples file can easily be replaced by an individual file. The key parameters would need some adjustments in this case.

Imports and Configuration

In [1]:
```
import numpy as np
import scipy.signal as signal
import scipy.io.wavfile as wav
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from matplotlib.ticker import EngFormatter
import sounddevice as sd
import os
import warnings

warnings.filterwarnings("ignore")

# ================================================ Matplotlib style
plt.rcParams.update({
    "figure.facecolor":  "#0d1117",
    "axes.facecolor":    "#161b22",
    "axes.edgecolor":    "#30363d",
    "axes.labelcolor":   "#c9d1d9",
    "xtick.color":       "#8b949e",
    "ytick.color":       "#8b949e",
    "text.color":        "#c9d1d9",
    "grid.color":        "#21262d",
    "grid.linestyle":    "--",
    "grid.alpha":        0.6,
    "lines.linewidth":   1.2,
    "font.family":       "monospace",
    "figure.dpi":        110,
})

ACCENT  = "#58a6ff"
ACCENT2 = "#3fb950"
ACCENT3 = "#f78166"
WARN    = "#e3b341"

# ============================ SDR / Signal parameters (edit these)
#
# 6 second recording
# - Center Frequency 144.5 MHz
# - Offset 30 kHz
# - 280 000 samples/s
```

```
# - 1.728e6 samples (6 times sample rate)
# - Gain 6 dB (close range)
#
# rtl_sdr -f 144470000 -s 280000 -d0 -g 6 -n 1728000 capture.iq
#
IQ_FILE         = "capture.iq"   # path to your raw IQ file
SAMPLE_RATE_IQ  = 280_000        # RTL-SDR sample rate (Hz)
CENTER_FREQ     = 144_470_000    # tuned centre frequency (Hz)
SIGNAL_FREQ     = 144_500_000    # actual signal frequency (Hz)

AUDIO_RATE      = 48_000         # output audio sample rate (Hz)
NFM_DEVIATION   = 2_500          # narrow FM deviation (Hz)
NFM_BANDWIDTH   = 8_500          # pre-demod IF filter bandwidth (Hz)
AUDIO_CUTOFF    = 3_400          # audio low-pass cutoff (Hz)
SQUELCH_DB      = -30            # squelch threshold (dBFS)

PLAY_AUDIO      = True           # set False to skip playback
SAVE_WAV        = True           # save demodulated audio to WAV
WAV_OUTPUT      = "output.wav"

print(f"[CONFIG]  IQ file      : {IQ_FILE}")
print(f"[CONFIG]  IQ rate      : {SAMPLE_RATE_IQ/1e3:.1f} kHz")
print(f"[CONFIG]  Centre freq  : {CENTER_FREQ/1e6:.4f} MHz")
print(f"[CONFIG]  Signal freq  : {SIGNAL_FREQ/1e6:.4f} MHz")
print(f"[CONFIG]  Audio rate   : {AUDIO_RATE/1e3:.1f} kHz")
```

```
[CONFIG]  IQ file      : capture.iq
[CONFIG]  IQ rate      : 280.0 kHz
[CONFIG]  Centre freq  : 144.4700 MHz
[CONFIG]  Signal freq  : 144.5000 MHz
[CONFIG]  Audio rate   : 48.0 kHz
```

Load IQ File

In [2]:

```
def load_iq(path: str) -> np.ndarray:
    """
    Load IQ samples from an RTL-SDR capture file.

    Format:

    uint8   : raw 8-bit unsigned (I,Q interleaved, offset binary)
    """
    raw = np.fromfile(path, dtype=np.uint8)

    # convert offset-binary uint8 -> float32 in [-1, 1]
    raw = (raw.astype(np.float32) - 127.5) / 127.5

    # de-interleave I and Q (iq is of type complex64)
    iq = raw[0::2] + 1j * raw[1::2]
    return iq


# ============================================================ Load
iq_data = load_iq(IQ_FILE)
duration = len(iq_data) / SAMPLE_RATE_IQ

print(f"[LOAD]  Samples     : {len(iq_data):,}")
print(f"[LOAD]  Duration    : {duration:.3f} s")
```

```
print(f"[LOAD]  I range     : [{iq_data.real.min():.4f}, {iq_data.real.ma
print(f"[LOAD]  Q range     : [{iq_data.imag.min():.4f}, {iq_data.imag.ma
```

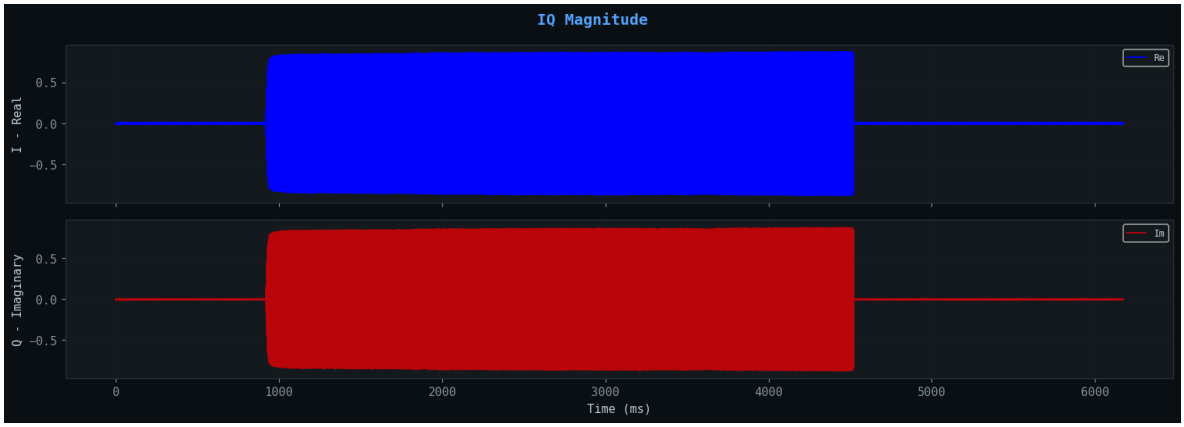
```
[LOAD]  Samples     : 1,728,000
[LOAD]  Duration    : 6.171 s
[LOAD]  I range     : [-0.8824, 0.8745]
[LOAD]  Q range     : [-0.8824, 0.8824]
```

IQ Magnitude

This output gives a quick overview about the recorded signal magnitude, separately for the In-Phase signal and the Quadrature signal.

This plot is not a necessary step for the demodulation process.

In [3]:
```
iq_mag_len = len(iq_data)
iq_mag_time = np.arange(iq_mag_len) / SAMPLE_RATE_IQ * 1e3   # millisecon
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 5), sharex=True)
fig.suptitle("IQ Magnitude", fontsize=13,
             color=ACCENT, fontweight="bold")
ax1.plot(iq_mag_time, np.real(iq_data), color="#0000FF", label='Re')
ax1.legend(fontsize=8)
ax1.set_ylabel('I - Real')
ax1.grid(True)
ax2.plot(iq_mag_time, np.imag(iq_data), color="#FF0000", alpha=0.7, label
ax2.legend(fontsize=8)
ax2.set_xlabel('Time (ms)')
ax2.set_ylabel('Q - Imaginary')
ax2.grid(True)
plt.tight_layout()
plt.savefig("plot_01_iq_magnitude.png", dpi=110, bbox_inches="tight")
plt.show()
print("[PLOT 1]  Saved -> plot_01_iq_magnitude.png")
```



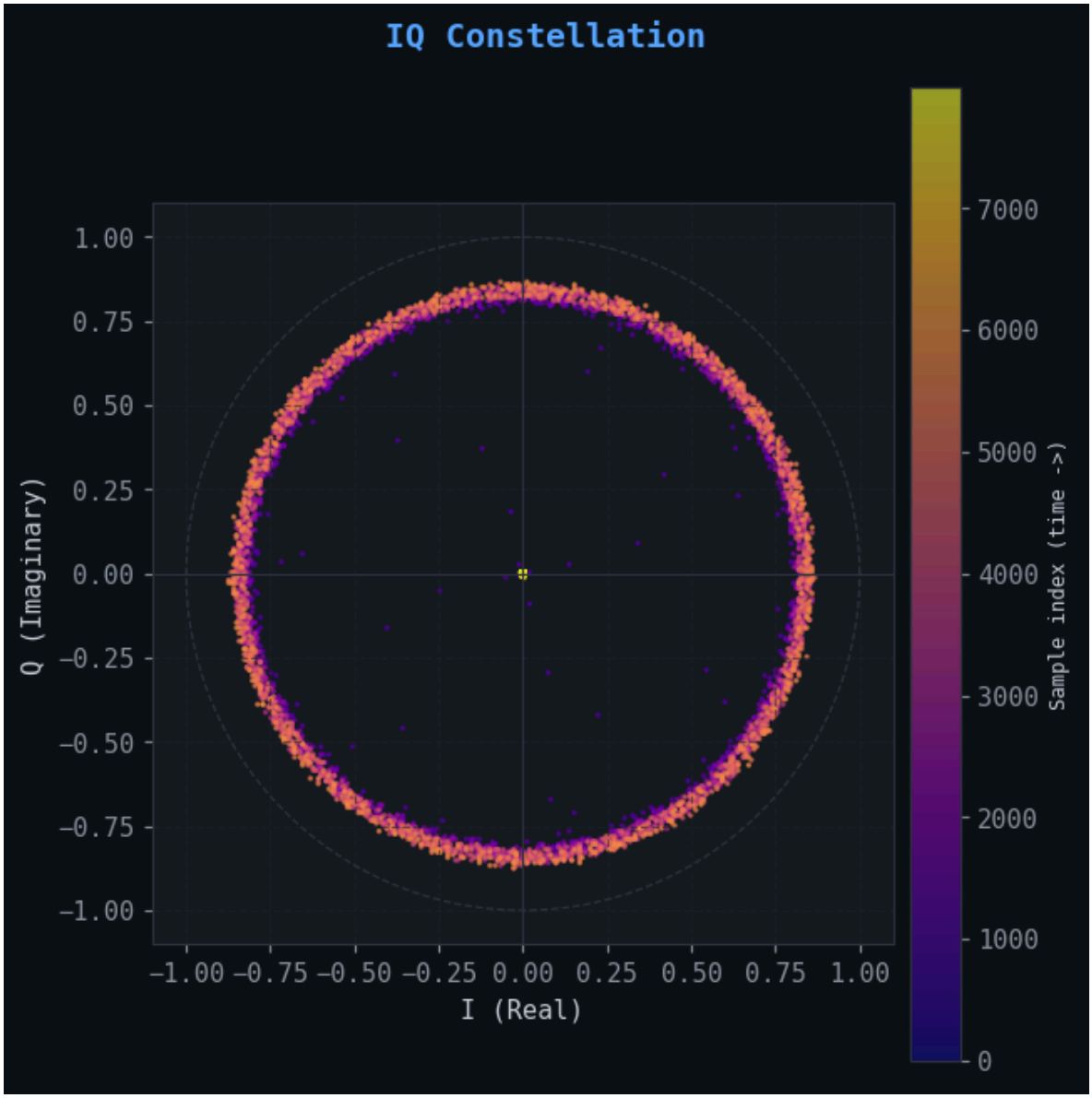

```
[PLOT 1]  Saved -> plot_01_iq_magnitude.png
```

IQ - Constellation

This is a representation of the first 8000 IQ samples in the complex plane. For FM-Modulation this should show a circle, more or less in the medium range.

This plot is not a necessary step for the demodulation process.

In [4]:
```
max_pts = 8000
# Step through the IQ data in 8000 equidistant steps
step = max(1, len(iq_data) // max_pts)
iq_sub = iq_data[::step][:max_pts]

fig, ax = plt.subplots(figsize=(6, 6))
fig.suptitle("IQ Constellation", fontsize=13,
             color=ACCENT, fontweight="bold")


scatter = ax.scatter(iq_sub.real, iq_sub.imag,
                     c=np.arange(len(iq_sub)), cmap="plasma",
                     s=1.5, alpha=0.6)

ax.axhline(0, color="#30363d", lw=0.8)
ax.axvline(0, color="#30363d", lw=0.8)
ax.set_xlabel("I (Real)")
ax.set_ylabel("Q (Imaginary)")
ax.set_aspect("equal")
ax.grid(True)

# unit circle reference
theta = np.linspace(0, 2 * np.pi, 360)
ax.plot(np.cos(theta), np.sin(theta), color="#30363d", lw=0.8, ls="--")

cbar = plt.colorbar(scatter, ax=ax, pad=0.02)
cbar.set_label("Sample index (time ->)", fontsize=8)

plt.tight_layout()
plt.savefig("plot_02_constellation.png", dpi=110, bbox_inches="tight")
plt.show()
print("[PLOT 2]  Saved -> plot_02_constellation.png")
```




```
[PLOT 2]  Saved -> plot_02_constellation.png
```

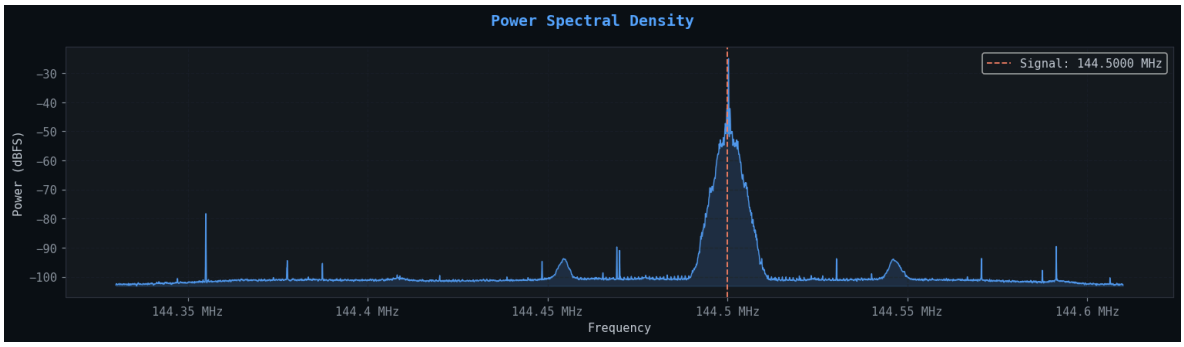
Power Spectral Density

The PSD graph shows the energy of signal in the frequency domain, as a spectrum. Here it can be checked if the expected signal frequency as defined in the key parameters ealier matches the power peak of the signal.

If not, the SIGNAL_FREQ parameter needs to be adjusted accordingly, otherwise the demodulation would not be optimal or fail entirely.

In [5]:
```
freq_fmt = EngFormatter(unit="Hz")

# Welch PSD of raw IQ\n"
f_psd, psd = signal.welch(iq_data, fs=SAMPLE_RATE_IQ, nperseg=4096, retur

f_psd = np.fft.fftshift(f_psd) + CENTER_FREQ
psd   = np.fft.fftshift(psd)
psd_db = 10 * np.log10(psd + 1e-20)

fig, ax = plt.subplots(figsize=(14, 4))
```

```
fig.suptitle("Power Spectral Density", fontsize=13, color=ACCENT, fontwei

ax.plot(f_psd, psd_db, color=ACCENT, lw=0.9, alpha=0.9)
ax.fill_between(f_psd, psd_db, psd_db.min(), alpha=0.15, color=ACCENT)
ax.axvline(SIGNAL_FREQ, color=ACCENT3, lw=1.2, ls="--",
           label=f"Signal: {SIGNAL_FREQ/1e6:.4f} MHz")

ax.set_xlabel("Frequency")
ax.set_ylabel("Power (dBFS)")
ax.xaxis.set_major_formatter(freq_fmt)
ax.grid(True)
ax.legend()

plt.tight_layout()
plt.savefig("plot_03_psd.png", dpi=110, bbox_inches="tight")
plt.show()
print("[PLOT 3]  Saved -> plot_03_psd.png")
```




```
[PLOT 3]  Saved -> plot_03_psd.png
```

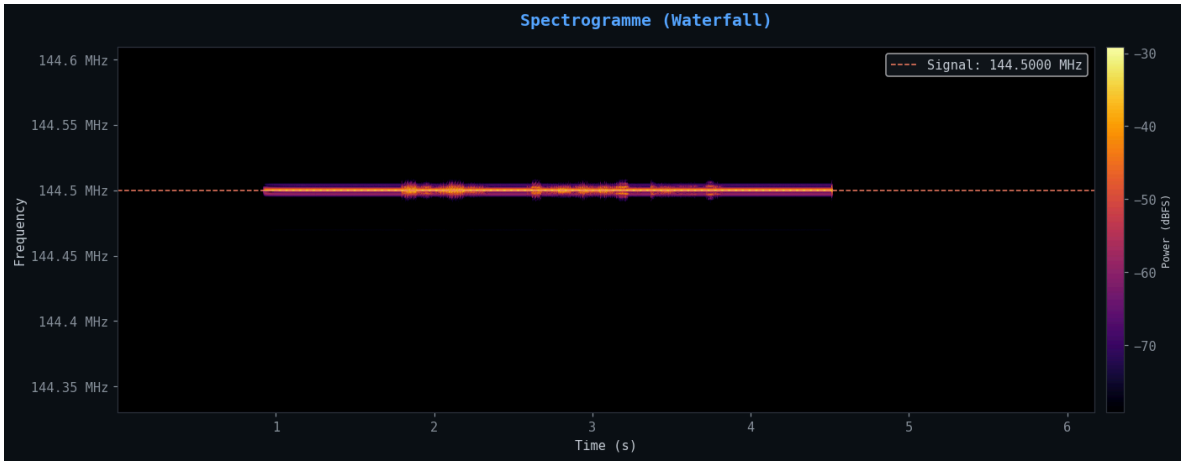
Spectrogramme

A fancy way to display the power ensity spectrum over time, again, not a necessary step in the FM Demodulation process, but a useful visual aid to inspect the signal.

In [6]:

```
fig, ax = plt.subplots(figsize=(14, 5))
fig.suptitle("Spectrogramme (Waterfall)", fontsize=13,
             color=ACCENT, fontweight="bold")

f_sg, t_sg, Sxx = signal.spectrogram(
    iq_data, fs=SAMPLE_RATE_IQ,
    nperseg=512, noverlap=384,
    return_onesided=False, mode="psd"
)
f_sg  = np.fft.fftshift(f_sg) + CENTER_FREQ
Sxx   = np.fft.fftshift(Sxx, axes=0)
Sxx_db = 10 * np.log10(Sxx + 1e-20)

im = ax.pcolormesh(t_sg, f_sg, Sxx_db,
                   shading="gouraud", cmap="inferno",
                   vmin=Sxx_db.max() - 50, vmax=Sxx_db.max())

ax.axhline(SIGNAL_FREQ, color=ACCENT3, lw=1.0, ls="--",
           label=f"Signal: {SIGNAL_FREQ/1e6:.4f} MHz")
ax.set_xlabel("Time (s)")
ax.set_ylabel("Frequency")
ax.yaxis.set_major_formatter(freq_fmt)
```

```
ax.legend(loc="upper right")

cbar = plt.colorbar(im, ax=ax, pad=0.01)
cbar.set_label("Power (dBFS)", fontsize=8)

plt.tight_layout()
plt.savefig("plot_04_waterfall.png", dpi=110, bbox_inches="tight")
plt.show()
print("[PLOT 4]  Saved -> plot_04_waterfall.png")
```




```
[PLOT 4]  Saved -> plot_04_waterfall.png
```

Frequency Shift to Baseband

This step is important when the signal frequency is not identical with the center frequency at which the IQ sample file was captured.

Otherwise the RTLSDR dongle alread moves the centerfrequency to the baseband at 0 Hz.

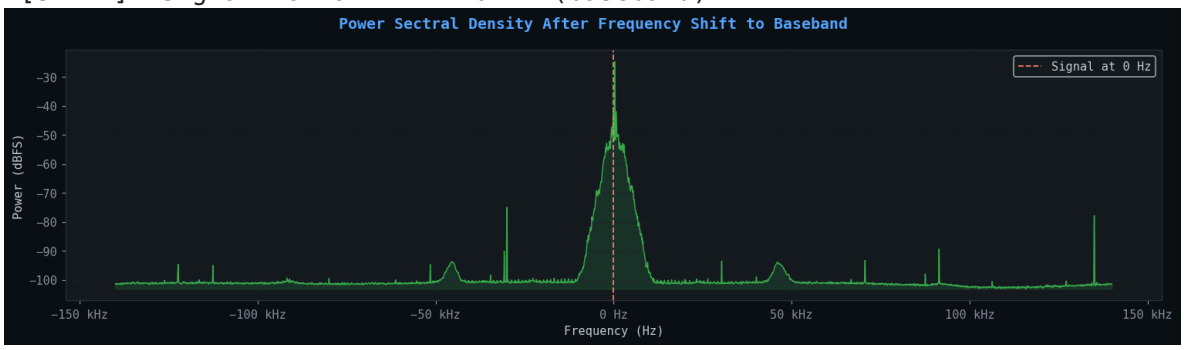
Negative frequencies don't make sense physically, but for the mathematical demodulation and filtering it simplifies the algorithms.

In [7]:

```
freq_offset = SIGNAL_FREQ - CENTER_FREQ

if abs(freq_offset) > 100:   # > 100 Hz threshold
    t_shift    = np.arange(len(iq_data)) / SAMPLE_RATE_IQ
    shift_vec  = np.exp(-1j * 2 * np.pi * freq_offset * t_shift).astype(n
    iq_shifted = iq_data * shift_vec
    print(f"[SHIFT]  Applied offset correction: {freq_offset:+,.0f} Hz")
else:
    iq_shifted = iq_data   # already at baseband
    print("[SHIFT]  Skipped — signal already at 0 Hz")

print(f"[SHIFT]  Frequency offset : {freq_offset:+,.0f} Hz")
print(f"[SHIFT]  Signal now at    :  0 Hz (baseband)")

# ========================================= Plot 5: PSD after shift
f_bb, psd_bb = signal.welch(
    iq_shifted, fs=SAMPLE_RATE_IQ, nperseg=4096, return_onesided=False
)
f_bb   = np.fft.fftshift(f_bb)
psd_bb = np.fft.fftshift(psd_bb)
```

```
fig, ax = plt.subplots(figsize=(14, 4))
fig.suptitle("Power Sectral Density After Frequency Shift to Baseband", f
             color=ACCENT, fontweight="bold")

ax.plot(f_bb, 10 * np.log10(psd_bb + 1e-20), color=ACCENT2, lw=0.9, alpha
ax.fill_between(f_bb, 10 * np.log10(psd_bb + 1e-20),
                10 * np.log10(psd_bb + 1e-20).min(),
                alpha=0.15, color=ACCENT2)
ax.axvline(0, color=ACCENT3, lw=1.2, ls="--", label="Signal at 0 Hz")
ax.set_xlabel("Frequency (Hz)")
ax.set_ylabel("Power (dBFS)")
ax.xaxis.set_major_formatter(EngFormatter(unit="Hz"))
ax.grid(True)
ax.legend()

plt.tight_layout()
plt.savefig("plot_05_shifted_psd.png", dpi=110, bbox_inches="tight")
plt.show()
print("[PLOT 5]  Saved -> plot_05_shifted_psd.png")
```

```
[SHIFT]  Applied offset correction: +30,000 Hz
[SHIFT]  Frequency offset : +30,000 Hz
[SHIFT]  Signal now at    :  0 Hz (baseband)
```




```
[PLOT 5]  Saved -> plot_05_shifted_psd.png
```

IF Band-Pass Filter

Cut off all signals out side of the interesting band around the signal frequency, before the demodulation.

In [8]:

```
if_bw       = NFM_BANDWIDTH

# The number of filter taps (coefficients) in the FIR filter.
# Must be odd for certain filters like band-pass.
taps        = 129 # filter order + 1
cut_off     = if_bw/2
if_lpf_b    = signal.firwin(taps, cut_off, fs=SAMPLE_RATE_IQ, window="ham

iq_filtered = signal.lfilter(if_lpf_b, 1.0, iq_shifted)

# ========================= Plot 6: Filter response + filtered PSD
f_resp, h_resp = signal.freqz(if_lpf_b, fs=SAMPLE_RATE_IQ, worN=8192, who
f_resp = f_resp - SAMPLE_RATE_IQ / 2
h_db   = 20 * np.log10(np.abs(np.fft.fftshift(h_resp)) + 1e-20)

f_filt, psd_filt = signal.welch(iq_filtered, fs=SAMPLE_RATE_IQ, nperseg=4
```

```
f_filt   = np.fft.fftshift(f_filt)
psd_filt = np.fft.fftshift(psd_filt)

fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 7), sharex=True)
fig.suptitle("IF Band-Pass Filter (NFM Channel)", fontsize=13,
             color=ACCENT, fontweight="bold")

ax1.plot(f_resp, h_db, color=WARN, lw=1.2, label="Filter response")
ax1.axvspan(-if_bw / 2, if_bw / 2, color=WARN, alpha=0.08, label=f"Passba
ax1.set_ylabel("Gain (dB)")
ax1.set_ylim(-180, 5)
ax1.grid(True)
ax1.legend()

ax2.plot(f_filt, 10 * np.log10(psd_filt + 1e-20), color=ACCENT2, lw=0.9,
ax2.fill_between(f_filt, 10 * np.log10(psd_filt + 1e-20),
                 10 * np.log10(psd_filt + 1e-20).min(),
                 alpha=0.15, color=ACCENT2)
ax2.axvspan(-if_bw / 2, if_bw / 2, color=WARN, alpha=0.06)
ax2.set_xlabel("Frequency (Hz)")
ax2.set_ylabel("Power (dBFS)")
ax2.xaxis.set_major_formatter(EngFormatter(unit="Hz"))
ax2.grid(True)

plt.tight_layout()
plt.savefig("plot_06_if_filter.png", dpi=110, bbox_inches="tight")
plt.show()
print(f"[FILTER]  IF filter taps : {taps}  bandwidth : {if_bw/1e3:.1f} kH
print("[PLOT 6]  Saved -> plot_06_if_filter.png")
```

```
[FILTER]  IF filter taps : 129  bandwidth : 8.5 kHz
[PLOT 6]  Saved -> plot_06_if_filter.png
```

FM - Discrimination

Continuous-time FM signal and demodulation

The FM signal has an instantaneous phase that encodes the modulated signal $m(\tau)$ as:

$$s(t) = A\cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\,d\tau\right)$$

The signal is recovered by differentiating the instantaneous phase:

$$m(t) = \frac{1}{2\pi k_f}\frac{d\phi(t)}{dt}$$

where $\phi(t)$ is the instantaneous phase and k_f is the frequency deviation constant (Hz/unit).

Discrete-time FM discriminator (as implemented)

For a complex baseband IQ sequence $x[n]$, the instantaneous phase difference between successive samples is:

$$\phi[n] = \angle(x[n] \cdot x^*[n-1])$$

The normalised demodulated audio sample is then:

$$y[n] = \frac{f_s}{2\pi\,\Delta f} \cdot \phi[n]$$

where f_s is the IQ sample rate (280 kHz) and Δf is the peak FM deviation (2 500 Hz). The factor

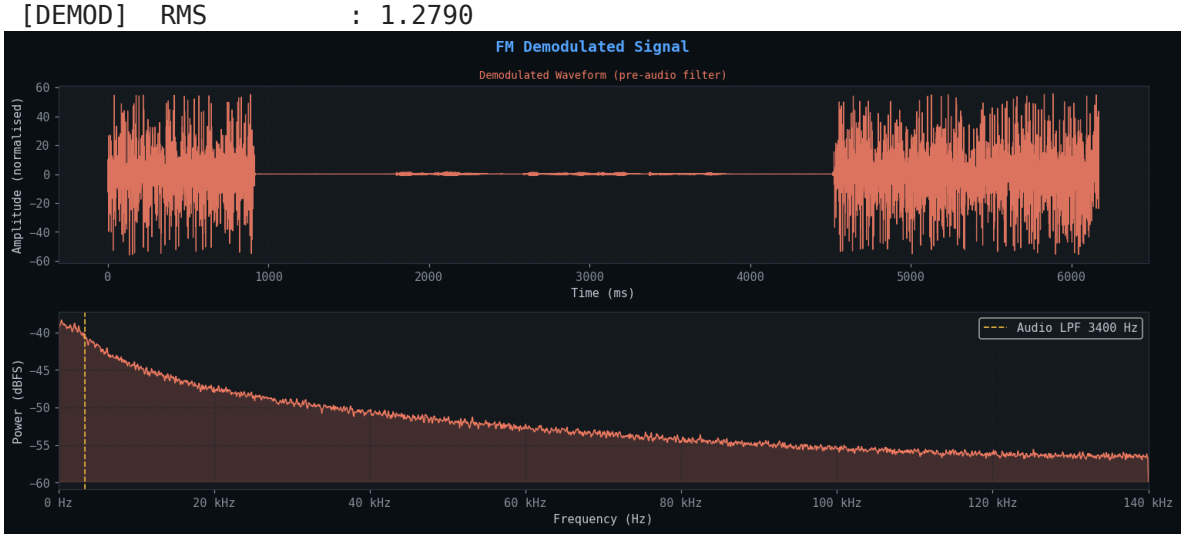
$$\frac{f_s}{2\pi\,\Delta f}$$

scales the output so that a full-deviation signal maps to ± 1.

In [9]:
```
def fm_demodulate(iq: np.ndarray, sample_rate: float,
                  deviation: float) -> np.ndarray:
    """
    FM discriminator using the instantaneous phase derivative.

    phi[n] = angle(iq[n] * conj(iq[n-1]))
    audio   = phi / (2*pi * deviation/fs)
    """
    delayed    = np.concatenate(([iq[0]], iq[:-1]))
    diff       = iq * np.conj(delayed)
    phase_diff = np.angle(diff)                        # radians per samp
    gain       = sample_rate / (2 * np.pi * deviation) # normalise to [-1
    return (phase_diff * gain).astype(np.float32)


demod = fm_demodulate(iq_filtered, SAMPLE_RATE_IQ, NFM_DEVIATION)
t_demod = np.arange(len(demod)) / SAMPLE_RATE_IQ

print(f"[DEMOD]  Samples     : {len(demod):,}")
print(f"[DEMOD]  Peak value  : {np.abs(demod).max():.4f}")
print(f"[DEMOD]  RMS         : {np.sqrt(np.mean(demod**2)):.4f}")
```

```
# ==================================== Plot 7: Demodulated waveform
disp = len(demod)

fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 6))
fig.suptitle("FM Demodulated Signal", fontsize=13,
             color=ACCENT, fontweight="bold")

ax1.plot(t_demod * 1e3, demod, color=ACCENT3, lw=0.8, alpha=0.9)
ax1.set_xlabel("Time (ms)")
ax1.set_ylabel("Amplitude (normalised)")
ax1.set_title("Demodulated Waveform (pre-audio filter)", color=ACCENT3, f
ax1.grid(True)

# PSD of demodulated signal
f_dm, psd_dm = signal.welch(demod, fs=SAMPLE_RATE_IQ,
                            nperseg=4096, return_onesided=True)
ax2.plot(f_dm, 10 * np.log10(psd_dm + 1e-20), color=ACCENT3, lw=0.9)
ax2.fill_between(f_dm, 10 * np.log10(psd_dm + 1e-20),
                 10 * np.log10(psd_dm + 1e-20).min(), alpha=0.2, color=AC
ax2.axvline(AUDIO_CUTOFF, color=WARN, ls="--", lw=1.2,
            label=f"Audio LPF {AUDIO_CUTOFF} Hz")
ax2.set_xlabel("Frequency (Hz)")
ax2.set_ylabel("Power (dBFS)")
ax2.set_xlim(0, SAMPLE_RATE_IQ / 2)
ax2.xaxis.set_major_formatter(EngFormatter(unit="Hz"))
ax2.legend()
ax2.grid(True)

plt.tight_layout()
plt.savefig("plot_07_demodulated.png", dpi=110, bbox_inches="tight")
plt.show()
print("[PLOT 7]  Saved -> plot_07_demodulated.png")
```

```
[DEMOD]  Samples     : 1,728,000
[DEMOD]  Peak value  : 55.9542
[DEMOD]  RMS         : 1.2790
```

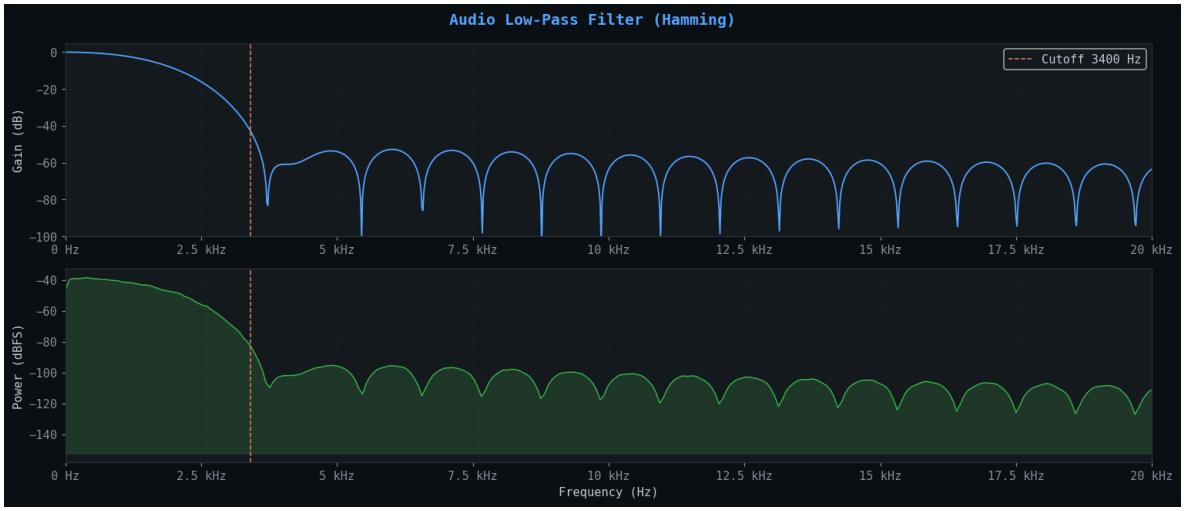



```
[PLOT 7]  Saved -> plot_07_demodulated.png
```

Audio Low-Pass Filter

In [10]:
```
audio_taps = 256
audio_lpf  = signal.firwin(audio_taps, AUDIO_CUTOFF / SAMPLE_RATE_IQ)
audio_if   = signal.lfilter(audio_lpf, 1.0, demod)
```

```
# ==================================== Plot 8: Audio filter response
f_ar, h_ar = signal.freqz(audio_lpf, fs=SAMPLE_RATE_IQ,
                          worN=8192, whole=False)
h_ar_db = 20 * np.log10(np.abs(h_ar) + 1e-20)

fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 6))
fig.suptitle("Audio Low-Pass Filter (Hamming)", fontsize=13,
             color=ACCENT, fontweight="bold")

ax1.plot(f_ar, h_ar_db, color=ACCENT, lw=1.2)
ax1.axvline(AUDIO_CUTOFF, color=ACCENT3, ls="--", lw=1.0,
            label=f"Cutoff {AUDIO_CUTOFF} Hz")
ax1.set_ylabel("Gain (dB)")
ax1.set_ylim(-100, 5)
ax1.set_xlim(0, 20_000)
ax1.xaxis.set_major_formatter(EngFormatter(unit="Hz"))
ax1.legend()
ax1.grid(True)

f_ai, psd_ai = signal.welch(audio_if, fs=SAMPLE_RATE_IQ,
                            nperseg=4096, return_onesided=True)
ax2.plot(f_ai, 10 * np.log10(psd_ai + 1e-20), color=ACCENT2, lw=0.9)
ax2.fill_between(f_ai, 10 * np.log10(psd_ai + 1e-20),
                 10 * np.log10(psd_ai + 1e-20).min(), alpha=0.2, color=AC
ax2.axvline(AUDIO_CUTOFF, color=ACCENT3, ls="--", lw=1.0)
ax2.set_xlabel("Frequency (Hz)")
ax2.set_ylabel("Power (dBFS)")
ax2.set_xlim(0, 20_000)
ax2.xaxis.set_major_formatter(EngFormatter(unit="Hz"))
ax2.grid(True)

plt.tight_layout()
plt.savefig("plot_08_audio_filter.png", dpi=110, bbox_inches="tight")
plt.show()
print("[PLOT 8]  Saved -> plot_08_audio_filter.png")
```



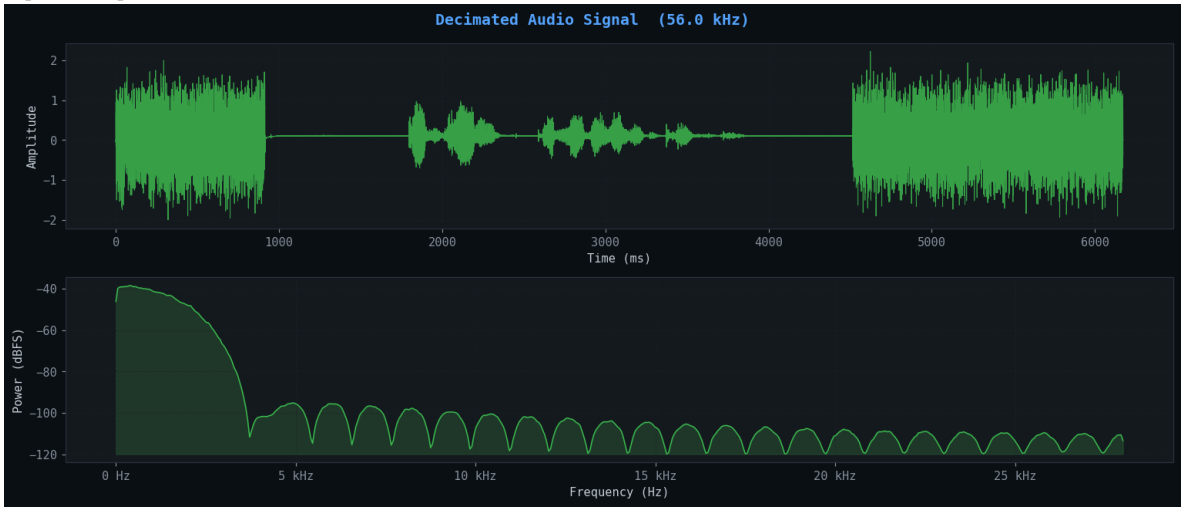

```
[PLOT 8]  Saved -> plot_08_audio_filter.png
```

Downsample to audio rate

In [11]:
```
decim_factor = int(SAMPLE_RATE_IQ / AUDIO_RATE)
if decim_factor < 1:
```

```
    decim_factor = 1

audio_decimated = audio_if[::decim_factor].copy()
actual_rate     = SAMPLE_RATE_IQ / decim_factor

print(f"[DECIM]  Factor      : {decim_factor}")
print(f"[DECIM]  Input rate  : {SAMPLE_RATE_IQ/1e3:.1f} kHz")
print(f"[DECIM]  Output rate : {actual_rate/1e3:.1f} kHz  (target {AUDIO_
print(f"[DECIM]  Samples     : {len(audio_decimated):,}")
print(f"[DECIM]  Duration    : {len(audio_decimated)/actual_rate:.3f} s")

# ================================ Plot 9: Decimated audio waveform
t_audio = np.arange(len(audio_decimated)) / actual_rate

fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 6))
fig.suptitle(f"Decimated Audio Signal  ({actual_rate/1e3:.1f} kHz)", font
             color=ACCENT, fontweight="bold")

ax1.plot(t_audio * 1e3, audio_decimated, color=ACCENT2, lw=0.7, alpha=0.8
ax1.set_xlabel("Time (ms)")
ax1.set_ylabel("Amplitude")
ax1.grid(True)

f_aud, psd_aud = signal.welch(audio_decimated, fs=actual_rate,
                              nperseg=1024, return_onesided=True)
ax2.plot(f_aud, 10 * np.log10(psd_aud + 1e-20), color=ACCENT2, lw=1.0)
ax2.fill_between(f_aud, 10 * np.log10(psd_aud + 1e-20),
                 10 * np.log10(psd_aud + 1e-20).min(), alpha=0.2, color=A
ax2.set_xlabel("Frequency (Hz)")
ax2.set_ylabel("Power (dBFS)")
ax2.xaxis.set_major_formatter(EngFormatter(unit="Hz"))
ax2.grid(True)

plt.tight_layout()
plt.savefig("plot_09_audio_decimated.png", dpi=110, bbox_inches="tight")
plt.show()
print("[PLOT 9]  Saved -> plot_09_audio_decimated.png")
```

```
[DECIM]  Factor      : 5
[DECIM]  Input rate  : 280.0 kHz
[DECIM]  Output rate : 56.0 kHz  (target 48.0 kHz)
[DECIM]  Samples     : 345,600
[DECIM]  Duration    : 6.171 s
```




```
[PLOT 9]  Saved -> plot_09_audio_decimated.png
```

Audio Squelch & Normalisation

Step 1

Suppress audio samples whenever the energy of the IQ signal is below the given threshold

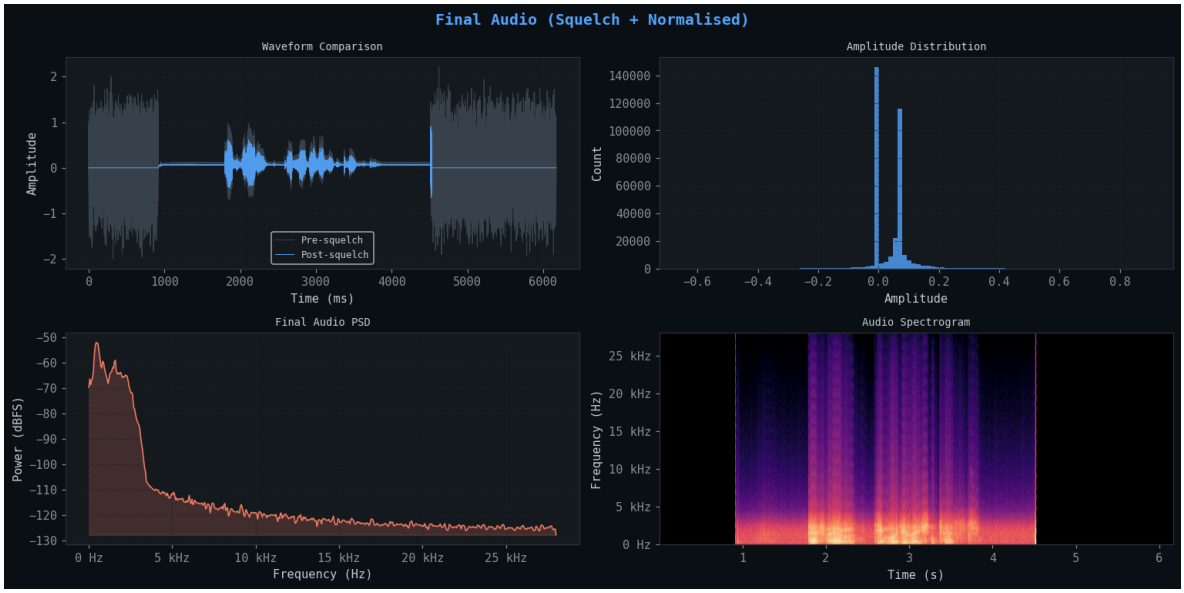
Step 2

Normalise the remaining audio signal to 90% of the maximum range.

In [12]:
```
def apply_squelch(audio: np.ndarray, iq_src: np.ndarray,
                  iq_rate: float, audio_rate: float,
                  threshold_db: float, block_ms: float = 10.0) -> np.ndar
    """
    Simple energy squelch.  Measures IQ signal power in blocks and
    zeros audio blocks below the threshold.
    """
    iq_block   = int(iq_rate * block_ms / 1000)
    audio_block = int(audio_rate * block_ms / 1000)
    out = audio.copy()

    for i in range(0, len(audio), audio_block):
        iq_start = int(i * iq_rate / audio_rate)
        iq_end   = iq_start + iq_block
        chunk    = iq_src[iq_start:iq_end]
        if len(chunk) == 0:
            break
        power_db = 10 * np.log10(np.mean(np.abs(chunk) ** 2) + 1e-20)
        if power_db < threshold_db:
            out[i:i + audio_block] = 0.0
    return out


audio_squelched = apply_squelch(
    audio_decimated, iq_data,
    SAMPLE_RATE_IQ, actual_rate,
    SQUELCH_DB
)

# ============================================ Normalise to [-1, 1]
peak = np.abs(audio_squelched).max()
if peak > 0:
    audio_norm = (audio_squelched / peak * 0.9).astype(np.float32)
else:
    audio_norm = audio_squelched.astype(np.float32)

print(f"[SQUELCH]  Threshold : {SQUELCH_DB} dBFS")
print(f"[NORM]     Peak      : {peak:.5f}  ->  normalised to 0.9")

# ================================= Plot 10: Final audio waveform
fig, axes = plt.subplots(2, 2, figsize=(14, 7))
fig.suptitle("Final Audio (Squelch + Normalised)", fontsize=13,
             color=ACCENT, fontweight="bold")
```

```
# Waveform comparison
axes[0, 0].plot(t_audio * 1e3, audio_decimated, color="#444c56", lw=0.7,
axes[0, 0].plot(t_audio * 1e3, audio_norm,      color=ACCENT,    lw=0.7,
axes[0, 0].set_xlabel("Time (ms)")
axes[0, 0].set_ylabel("Amplitude")
axes[0, 0].set_title("Waveform Comparison", fontsize=9)
axes[0, 0].legend(fontsize=8)
axes[0, 0].grid(True)

# Amplitude histogram
axes[0, 1].hist(audio_norm, bins=100, color=ACCENT, alpha=0.8, edgecolor=
axes[0, 1].set_xlabel("Amplitude")
axes[0, 1].set_ylabel("Count")
axes[0, 1].set_title("Amplitude Distribution", fontsize=9)
axes[0, 1].grid(True)

# Final PSD
f_fn, psd_fn = signal.welch(audio_norm, fs=actual_rate,
                            nperseg=1024, return_onesided=True)
axes[1, 0].plot(f_fn, 10 * np.log10(psd_fn + 1e-20), color=ACCENT3, lw=1.
axes[1, 0].fill_between(f_fn, 10 * np.log10(psd_fn + 1e-20),
                        10 * np.log10(psd_fn + 1e-20).min(), alpha=0.2,
axes[1, 0].set_xlabel("Frequency (Hz)")
axes[1, 0].set_ylabel("Power (dBFS)")
axes[1, 0].set_title("Final Audio PSD", fontsize=9)
axes[1, 0].xaxis.set_major_formatter(EngFormatter(unit="Hz"))
axes[1, 0].grid(True)

# Spectrogram of final audio
f_sp, t_sp, Sxx_a = signal.spectrogram(
    audio_norm, fs=actual_rate, nperseg=256, noverlap=200
)
axes[1, 1].pcolormesh(t_sp, f_sp, 10 * np.log10(Sxx_a + 1e-20),
                      shading="gouraud", cmap="magma")
axes[1, 1].set_xlabel("Time (s)")
axes[1, 1].set_ylabel("Frequency (Hz)")
axes[1, 1].set_title("Audio Spectrogram", fontsize=9)
axes[1, 1].yaxis.set_major_formatter(EngFormatter(unit="Hz"))

plt.tight_layout()
plt.savefig("plot_10_final_audio.png", dpi=110, bbox_inches="tight")
plt.show()
print("[PLOT 10]  Saved -> plot_10_final_audio.png")
```

```
[SQUELCH]  Threshold : -30 dBFS
[NORM]     Peak      : 1.43515  ->  normalised to 0.9
```




```
[PLOT 10]  Saved -> plot_10_final_audio.png
```

Save WAV file and play audio

In [13]:
```
# ======================================================= Save to WAV
if SAVE_WAV:
    wav_data = (audio_norm * 32767).astype(np.int16)
    wav.write(WAV_OUTPUT, int(actual_rate), wav_data)
    size_kb = os.path.getsize(WAV_OUTPUT) / 1024
    print(f"[WAV]   Saved  ->  {WAV_OUTPUT}  ({size_kb:.1f} KB)")
    print(f"[WAV]   Rate   : {actual_rate:.0f} Hz")
    print(f"[WAV]   Length : {len(wav_data)/actual_rate:.3f} s")

# ======================================================= Playback
if PLAY_AUDIO:
    print(f"\n[PLAY]  Playing {len(audio_norm)/actual_rate:.2f} s of audi
          f"@ {actual_rate:.0f} Hz ...")
    sd.play(audio_norm, samplerate=int(actual_rate), blocking=True)
    sd.wait()
    print("[PLAY]  Done.")
else:
    print("[PLAY]  Playback disabled (PLAY_AUDIO = False)")
```

```
[WAV]   Saved  ->  output.wav  (675.0 KB)
[WAV]   Rate   : 56000 Hz
[WAV]   Length : 6.171 s

[PLAY]  Playing 6.17 s of audio @ 56000 Hz ...
[PLAY]  Done.
```

Summary Report

In [14]:
```
sep = "-" * 52

print(sep)
print("  RTL-SDR NARROW FM DEMODULATOR  —  SUMMARY")
print(sep)
print(f"  IQ file          : {IQ_FILE}")
print(f"  IQ sample rate   : {SAMPLE_RATE_IQ/1e3:.1f} kHz")
```

```
print(f"  Centre frequency : {CENTER_FREQ/1e6:.4f} MHz")
print(f"  Signal frequency : {SIGNAL_FREQ/1e6:.4f} MHz")
print(f"  Freq offset      : {freq_offset:+,.0f} Hz")
print(sep)
print(f"  IF BW (channel)  : {NFM_BANDWIDTH/1e3:.1f} kHz")
print(f"  FM deviation     : {NFM_DEVIATION/1e3:.2f} kHz")
print(f"  Audio LPF        : {AUDIO_CUTOFF} Hz")
print(f"  Decimation       : x{decim_factor}  ({SAMPLE_RATE_IQ/1e3:.0f} -
print(f"  Squelch          : {SQUELCH_DB} dBFS")
print(sep)
print(f"  Audio duration   : {len(audio_norm)/actual_rate:.3f} s")
print(f"  Peak amplitude   : {np.abs(audio_norm).max():.4f}")
print(f"  RMS amplitude    : {np.sqrt(np.mean(audio_norm**2)):.4f}")
print(sep)
print("  Plots generated:")
plots = [
    "plot_01_raw_iq.png          — Raw IQ time domain (I, Q, magnitude)",
    "plot_02_constellation.png   — IQ constellation diagram",
    "plot_03_psd.png             — Full-span PSD (Welch)",
    "plot_04_waterfall.png       — Spectrogram / waterfall",
    "plot_05_shifted_psd.png     — PSD after frequency shift",
    "plot_06_if_filter.png       — IF band-pass filter response",
    "plot_07_demodulated.png     — FM discriminator output",
    "plot_08_audio_filter.png    — Audio LPF response",
    "plot_09_audio_decimated.png — Decimated audio waveform",
    "plot_10_final_audio.png     — Final audio (4-panel)",
]
for p in plots:
    print(f"    {p}")
print(sep)
```

```
----------------------------------------------------
  RTL-SDR NARROW FM DEMODULATOR  —  SUMMARY
----------------------------------------------------
  IQ file          : capture.iq
  IQ sample rate   : 280.0 kHz
  Centre frequency : 144.4700 MHz
  Signal frequency : 144.5000 MHz
  Freq offset      : +30,000 Hz
----------------------------------------------------
  IF BW (channel)  : 8.5 kHz
  FM deviation     : 2.50 kHz
  Audio LPF        : 3400 Hz
  Decimation       : x5  (280 -> 56.0 kHz)
  Squelch          : -30 dBFS
----------------------------------------------------
  Audio duration   : 6.171 s
  Peak amplitude   : 0.9000
  RMS amplitude    : 0.0713
----------------------------------------------------
  Plots generated:
    plot_01_raw_iq.png          — Raw IQ time domain (I, Q, magnitude)
    plot_02_constellation.png   — IQ constellation diagram
    plot_03_psd.png             — Full-span PSD (Welch)
    plot_04_waterfall.png       — Spectrogram / waterfall
    plot_05_shifted_psd.png     — PSD after frequency shift
    plot_06_if_filter.png       — IF band-pass filter response
    plot_07_demodulated.png     — FM discriminator output
    plot_08_audio_filter.png    — Audio LPF response
    plot_09_audio_decimated.png — Decimated audio waveform
    plot_10_final_audio.png     — Final audio (4-panel)
----------------------------------------------------
```